

Analysis First-Year Exam, May 2016, Part 1

Do exactly four problems. Work should be presented in a logical fashion in order to receive full credit.

1. Let f be a continuous function on $[0, 1]$. Prove that f is uniformly continuous.
2. Let $\{G_n\}_{n=1}^{\infty}$ be a family of open sets in $\mathbb{R}^n$ and $\{F_n\}_{n=1}^{\infty}$ a family of closed sets. Prove that $\bigcup_{n=1}^{\infty} G_n$ is open and $\bigcup_{n=1}^N F_n$ is closed for any finite N . Give an example such that $\bigcup_{n=1}^{\infty} F_n$ is not closed.
3. Suppose $\sum_{n=1}^{\infty} a_n$ is an absolutely convergent real series. Let b_n be a re-arrangement of a_n . Prove that $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n$.
4. Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } x = y = 0. \end{cases}$$

Prove that f is not continuous at $(0, 0)$.

5. Suppose $f'(x) > 0$ for $x \in (0, 1)$ and let g be the inverse function of f . Prove that for $x_0 \in (0, 1)$, $g'(f(x_0)) = 1/f'(x_0)$.